

Challenge (Habanero)

Q1. Solve for x : $2^x = 32$

- (A) $x = 3$
- (B) $x = 4$
- (C) $x = 5$
- (D) $x = 6$
- (E) $x = 7$

Q2. A gaming platform's user base grows according to $y = 3^x$, where y is the number of users and x is the number of years. If $y = 81$, what is x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q3. A coder's score S is calculated by $S = 2^{n/2}$, where n is the number of levels completed. If $S = 4$, what is n ?

- (A) $n = 2$
- (B) $n = 4$
- (C) $n = 6$
- (D) $n = 8$
- (E) $n = 10$

Q4. Solve for x : $5^x = 125$

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q5. A data plan's cost C is given by $C = 2^{x+1}$, where x is the number of gigabytes used. If $C = 16$, what is x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q6. A gamer's ranking R is determined by $R = 3^{x-1}$, where x is the number of wins. If $R = 9$, what is x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q7. Solve for x : $4^x = 256$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q8. A software company's revenue R grows according to $R = 2^{x+2}$, where x is the number of years. If $R = 64$, what is x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Open Ended Questions (FRQ)

Q9. A gaming tournament's prize money P is given by $P = 5^{x-1}$, where x is the number of rounds won. If $P = 125$, find the value of x . Show your work and explain your reasoning.

Q10. A coder's salary S is calculated by $S = 3^{x/2}$, where x is the number of years of experience. If $S = 27$, find the value of x . Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Ensure students understand the properties of exponents
- Tip 2: Encourage students to use logarithms to solve exponential equations
- Tip 3: Remind students to check their work and explain their reasoning